



REST CODER ACADEMY

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JAVA PROGRAMMING QUESTIONS

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PATTERN PROGRAMMING

Develop a java program to print the below patterns.

1.

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

2.

```
11 12 13 14 15  
21 22 23 24 25  
31 32 33 34 35  
41 42 43 44 45  
51 52 53 54 55
```

3.

```
1 1 1 1 1  
2 2 2 2 2  
3 3 3 3 3  
4 4 4 4 4  
5 5 5 5 5
```

4.

```
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5
```

5.

```

*
 *
  *
   *
    *

```

6.

```

1
 2
   3
    4
     5

```

7.

```

* * * * *
 * * * * *
  * * * *
   * * *
    * *
     *

```

8.

```

1 1 1 1 1
 2 2 2 2
   3 3 3
    4 4
     5

```

9.

```

1 2 3 4 5
 2 3 4 5
   3 4 5
    4 5
     5

```

10.

```
*  
* *  
* * *  
* * * *  
* * * * *
```

11.

```
1  
2 2  
3 3 3  
4 4 4 4  
5 5 5 5 5
```

12.

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

13.

```
      *  
     *  
    *  
   *  
  *  
 *
```

14.

```
      *  
     * *  
    * * *  
   * * * *  
  * * * *  
 * * * * *
```

15.

```

      1
     2 2
    3 3 3
   4 4 4 4
  5 5 5 5 5

```

16.

```

      5
     4 5
    3 4 5
   2 3 4 5
  1 2 3 4 5

```

17.

```

* * * * *
* * * *
* * *
* *
*

```

18.

```

1 1 1 1 1
2 2 2 2
3 3 3
4 4
5

```

19.

```

1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

```

20.

1

2 3

4 5 6

7 8 9 10

21.

A

B C

D E F

G H I J

22.

A

B B

C C C

D D D D

23.

A	B	C	D
	E	F	G
		H	I
			J

24.

```
* * * * *
*           *
*           *
*           *
*           *
* * * * *
```

25.

* * * * *

* * * * *

* * * * *

26.

* * *

* * *

* * *

* * *

* * *

27.

* *

* *

* *

* *

* *

28.

* * * * *

* *

* *

* *

* * * * *

29.

* *

* * *

* * *

* * *

* *

30.

```

* * * * *
*           *
*           *
* * * * *
*           *
*           *
*           *

```

31.

```

* * * * *
*
*
* * * * *
           *
           *
* * * * *

```

32.

```

* * * * *
*           *
* * * * *
*
*

```

33.

```

      *
    * * *
  * * * * *
* * * * *

```

34.

```
      1
     1 2 3
    1 2 3 4 5
   1 2 3 4 5 6 7
```

35.

```
      1
     2 2 2
    3 3 3 3 3
   4 4 4 4 4 4 4
```

36.

```
      *
     * * *
    * * * * *
   * * * * * *
  * * * * *
 * * * *
  * * *
   * * *
    * * *
```



NUMBER SERIES / NUMBER SYSTEM

1. Develop a java program to check if a number is positive or negative.
2. Develop a java program to check if a number is even or odd.
3. Develop a java program to find largest of 2 numbers.
4. Develop a java program to find largest of 2 numbers using ternary operator.
5. Develop a java program to find largest of 3 numbers.
6. Develop a java program to find largest of 4 numbers.
7. Develop a java program to swap 2 numbers.
8. Develop a java program to swap 2 numbers without using third variable.
9. Develop a java program to find the sum of n natural numbers.

10. Develop a java program to find the sum of 'n' even natural numbers.

11. Develop a java program to find the sum of 'n' odd natural numbers.

12. Develop a java program to find the square and cube of a specific number.

13. Develop a java program to find the factorial of a number.

Input: 5

Output: 120

14. Develop a java program to find the factorial of n natural numbers.

Input: 6

Output:

1! is 1

2! is 2

3! is 6

4! is 24

5! is 120

6! is 720

15. Develop a java program to check if a number is prime or not.

16. Develop a java program to generate fibonacci series.

17. Develop a java program to get last digit from given number.

Input: 123

Output: 3

18. Develop a java program to get first digit from given number.

Input: 245

Output: 2

19. Develop a java program to reverse a number.

Input: 231

Output: 132

20. Develop a java program to check if a number is palindrome.

21. Develop a java program check if a number is Armstrong or not

22. Develop a java program to print all Armstrong numbers between 1 to 1000.

23. Develop a java program to check if a number is a Perfect Number(Example: $6 \rightarrow 1 + 2 + 3 = 6$)

24. Develop a java program Check if a number is a Strong Number. (Example: $145 \rightarrow 1! + 4! + 5! = 145$)

25. Develop a java program check if a number is a Harshad / Niven number.

26. Develop a java program to find the sum of digits of a number

27. Develop a java program to find the product of digits of a number

28. Develop a java program to count number of digits in a number

29. Develop a java program to find the largest digit in a number

30. Develop a java program to find the smallest digit in a number

RECURSION

1. Develop a java program to print the sequence using recursion.

- I. 1 2 3 4 5
- II. 5 4 3 2 1
- III. 2 4 6 8 10
- IV. 1 3 5 7 9
- V. 3 6 9 12 15 18 21 24 27 30
- VI. 10 8 6 4 2
- VII. 9 7 5 3 1

2. Develop a java program to find the sum of n natural numbers using recursion.

3. Develop a java program to find the sum of n even natural numbers using recursion.

4. Develop a java program to find the odd of n even natural numbers using recursion.

5. Develop a java program to find the factorial of a number using recursion.

6. Develop a java program to print fibonacci series using recursion.

7. Develop a java program to print the below sequence using recursion

- i. 3 2 1 0 1 2 3.**
- ii. 10 8 6 4 2 0 2 4 6 8 10.**
- iii. 10 8 6 4 2 0 1 3 5 7 9.**



ARRAY

1. Develop a java program to traverse array elements in forward & backward direction.
2. Develop a java program to find sum & average of array elements.
3. Develop a java program to print even numbers from given array.
4. Develop a java program to print odd numbers from given array.
5. Develop a java program to find no of even numbers from given array.
6. Develop a java program to find no of odd numbers from given array.
7. Develop a java program to find sum & average of even numbers from the array.
8. Develop a java program to find sum & average of odd numbers from the array.

9. Develop a java program to find largest number from the array.

10. Develop a java program to find smallest number from the array.

11. Develop a java program to find the number of occurrences of a specific element in an array.

12. Develop a java program to check if the element is present in the array or not (Linear Search).

13. Develop a java program to find the factorial of every element in the array.

14. Develop a java program to check if the element is present in the array or not (Binary Search).

15. Develop a Java program to sort the array in ascending order.

16. Develop a Java program to sort the array in descending order.

17. Develop a Java program to sort the array without using built-in methods. (*Bubble Sort*)

18. Develop a Java program to find duplicate elements in an array.

19. Develop a Java program to print unique elements of an array.

20. Develop a Java program to count frequency of each element in the array.

21. Develop a Java program to find the second largest element in an array.

22. Develop a Java program to find the second smallest element in an array.

23. Develop a Java program to find common elements between two arrays.

24. Develop a Java program to shift all zeros to the end of the array.

25. Develop a Java program to print a 2D matrix.

26. Develop a Java program to find the transpose of a matrix.

27. Develop a Java program to find the sum of each column in a 2D matrix.

28. Develop a Java program to find the sum of each row in a 2D matrix.

29. Develop a Java program to multiply two 2D matrices.



STRING

1. Write a java program to

- Print the String.
- Find length of the String.
- Compare 2 Strings are same or not.
- Concatenate 2 Strings.
- Remove leading and trailing whitespace.

Input: " java program "

Output: "java program"

- Count no of words in the given String.

Input: "This is java programming class";

Output: 5

2. Write a java program to reverse a string.

3. Develop a java program to check if a String is palindrome or not.

Input: MADAM

Output: It is a Palindrome

Input: ABC

Output: Not a Palindrome

4. Write a java program to count Vowels, Consonants & Others.

7. Write a Java program to count the frequency of each character in a string.

Example:

Input → "hello"

Output → h=1, e=1, l=2, o=1

8. Write a Java program to check if two strings are Anagrams.

Example:

Input → "listen", "silent"

Output → Anagram

9. Write a Java program to remove all spaces from a string.

Input → "a b c d"

Output → "abcd"

10. Write a Java program to replace all occurrences of a character in a string.

Input → "banana" replace 'a' → 'o'

Output → "bonono"

11. Write a Java program to convert a string to uppercase & lowercase.

Input → "Java"

Output → "JAVA" and "java"

12. Write a Java program to find the largest and smallest word in a sentence.

Input → "Java programming is easy"

Output → Largest = programming, Smallest = is

13. Write a Java program to count digits, letters & special characters in a string.

Input → "abc123@#"

Output → letters=3, digits=3, special=2

14. Write a Java program to reverse the words in a sentence.

Input → "Java is fun"

Output → "fun is Java"

15. Write a Java program to find duplicate characters in a string.

Input → "programming"

Output → r g m (characters that repeat)

16. Write a Java program to remove duplicate characters from a string.

Input → "banana"

Output → "ban"

17. Write a Java program to swap two strings without using a third variable.

Input → "Java", "Python"

Output → swapped values

18. Write a Java program to count how many times a specific character appears in a string.

Example:

Input → "banana", character → 'a'

Output → 3

19. Write a Java program to check whether a string contains a specific substring.

Example:

Input → "Learn Java Programming", substring → "Java"

Output → Present



COLLECTION

1. Write a Java program to remove duplicate elements from an ArrayList using a Set.
2. Write a Java program to find the second largest element in an ArrayList of integers.
3. Write a Java program to count the frequency of each element in an ArrayList using HashMap.
Input: [2,3,2,5,3]
Output: {2=2, 3=2, 5=1}
4. Write a Java program to find common elements between two ArrayLists.
Input: [1,2,3,4] & [3,4,5]
Output: [3,4]
5. Write a Java program to find unique elements from an ArrayList.
Input: [2,2,3,4,3]
Output: [4]
6. Write a Java program to remove all even numbers from an ArrayList.
Input: [1,2,3,4,5,6]
Output: [1,3,5]

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